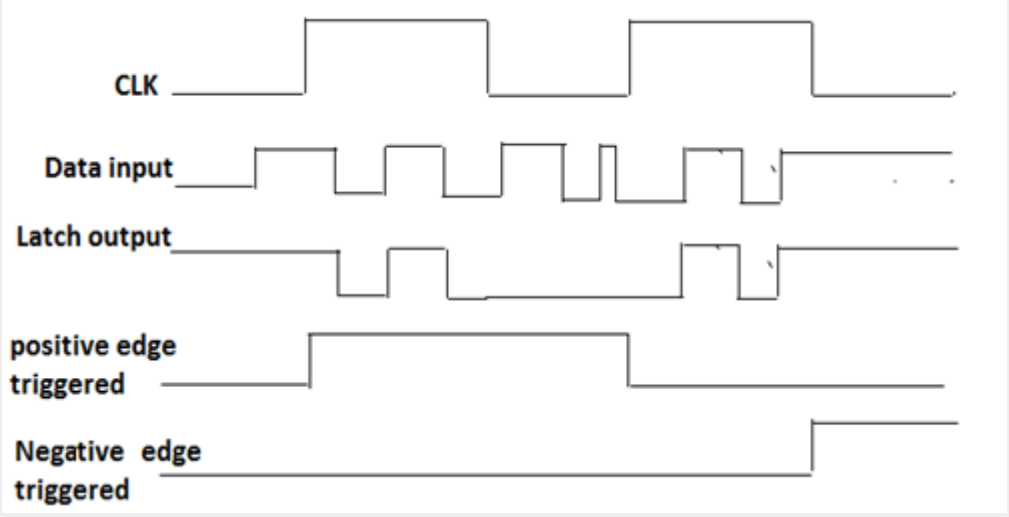
DIGITAL FAQ'S

1. What are the applications of buffer?

Ans : To introduce small delays. To eliminate cross talk caused due to inter electrode capacitance due to close routing, buffers support highfan-out.

2. For following data signal draw output waveform for gated d latch positive edge triggered and negative edge triggered d flipflop?

Ans :



3. Why Digital signal are mostly used for transmission than analog?

Ans : Digital Signals are easy to store and easy to manipulate without error. Accuracy increases by increasing number of levels. Here cost of thehardware is proportional to accuracy. Where in analog, signal values are stored in capacitors it is not constant for long time it may increase or decrease because of noise. Accuracy increases by increasing more sensible coils. Here the cost of the require hardware is much more to α to accuracy.

4. What is excitation table?

Ans : Minimum number of inputs that are necessary to generate a particular next state when the current state is known. Similar to truth tableor state table, but rearrange the data so that the current state and next state are next to each other on the L.H.S of the table and the inputs needed to make that state change happened are shown on the R.H.S of the table.

Characteristic Table

S	R	ACTION
0	0	Keep State
0	1	Q = 0
1	0	Q = 1
1	1	Race Condition

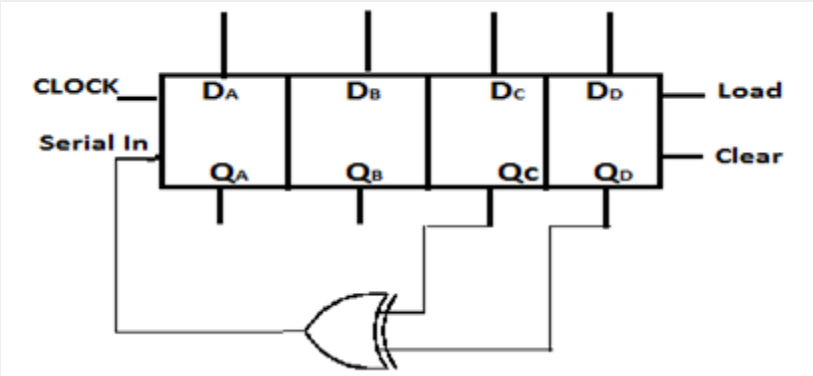
Q(t)	Q(t+1)	S	R
0	0	0	X
1	0	0	1
0	1	1	0
1	1	X	1

5. What is difference between synchronous flip-flop and asynchronous flip-flop?

Ans : In asynchronous changing state bits are used as clock to subsequent state machines. Synchronous all flip flops gets clock at the same time.All state bits change under the control of single clock.

6. Draw pseudo random sequent sequence generator to generate 15 random patterns?

Ans : Sequence depends on initial pattern. XOR gate can be given at any where.



7. What are the applications of pseudo random sequent sequence generator?

Ans : To check transmission over a wire for period of time. To Establishing the reliability communication between two points. To test anysystem.

8. How to find the Propagation delay, Clock to Output?

Ans : The summation of all delays encountered from where the clock occurs to the output. In short, the delays of the State memory (R) and theoutput logic (G).
PD Clock- Output (min) = Rpd (min) + Gpd (min)
PD Clock- Output (max) = Rpd (max) + Gpd (max)

9. How to find the Propagation delay, input to output delay?

Ans : This is a property associated with Mealy machines only. The calculation is the summation of all propagation delays encountered betweenthe input and the output.
For MOORE machines:
PD Input- Output (min) = infinity (∞)
PD Input- Output (max) = infinity (∞)
For MEALY Machines:
PD Input- Output (min) = Gpd (min)
PD Input- Output (max) = Gpd (max)

10. How to calculate Setup time?

Ans : The calculation for setup time is the sum of the setup time for the concerned flip flop and the maximum delay from the input logic (F).
T SETUP = RSetup+ Fpd (MAX).

11. How to calculate the value for Hold time?

Ans : T HOLD = RHold - Fpd (MIN)

12. How to calculate the Maximum Clock rate (MCLK)?

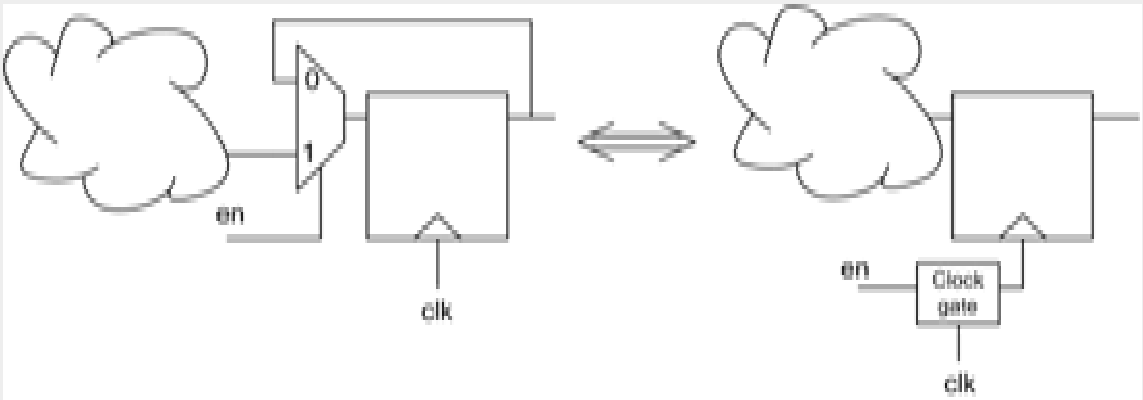
Ans : MCLK = 1/ TMIN Calculate TMIN first, TMIN refers to the minimum time period for correct operation of the circuit, it is calculated using allworst cases i.e maximum delays.
TMIN = Fpd (MAX) + RSetup + Rpd (MAX)

13. What is Clock Skew?

Ans : Given two sequentially-adjacent registers, Ri and Rj and an equipotential clock distribution network, the clock skew between these tworegisters is defined as
Tskew = Tci - Tcj
Here Tci and Tcj are the clock delays from the clock source to the registers Ri and Rj, respectively.

14. Define Clock Gating?

Ans : Clock gating is one of the power-saving techniques used on the Pentium 4 processor. To save power, clock gating refers to activating theclocks in a logic block only when there is work to be done.



15. Why NAND gate is preferable over NOR gate for fabrication?

Ans : In NAND gates at the transistor level the mobility of electrons is normally three times compared to holes mobility in NOR, thus the NAND isa faster gate. Additionally, the gate-leakage in NAND structures is much lower.

16. What is Noise Margin?

Ans : The minimum amount of noise that can be allowed on the input stage for which the output won't effect.

17. When metastability will occur? Different ways to avoid this?

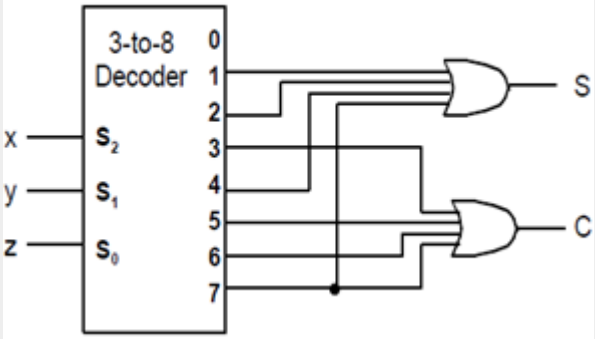
Ans : This will happen if the O/P cap is not allowed to charge/discharge fully to the required logical levels. If there is a setup time violation,metastability will occur, to avoid this, a series of flip-flops is used (normally 2 or 3) which will remove the intermediate states.

18. For a NAND gate if signal A arrives at the NAND gate later than signal B. To optimize delay of the two series inputs A and B which one is placed near to the output?

Ans : The late coming signals are to be placed closer to the output node i.e, A should be placed closer to the output.

19. How to implement Full adder using decoder?

Ans : S(x, y, z) = S (1, 2, 4, 7)
C(x, y, z) = S (3, 5, 6, 7)



20. Given only two xor gates one must function as buffer and another as inverter?

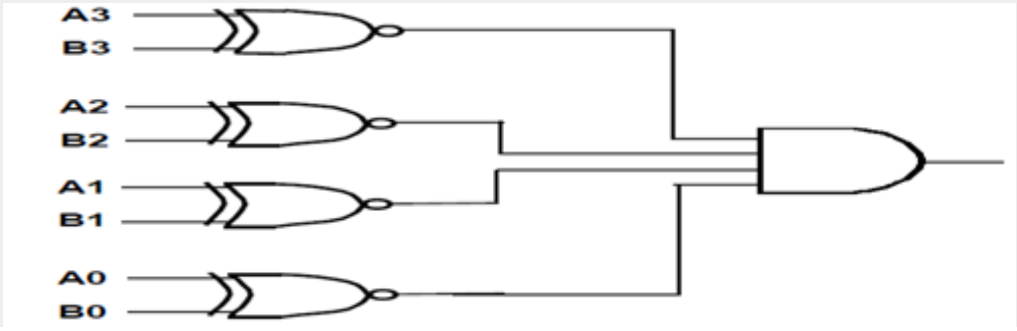
Ans : Tie one of xor gates input to 1 it will act as inverter. Tie one of xor gates input to 0 it will act as buffer.

21. How to achieve 180 degree exact phase shift?

Ans : Bufgds that is differential signaling buffers which are inbuilt resource of most of FPGA can be used.

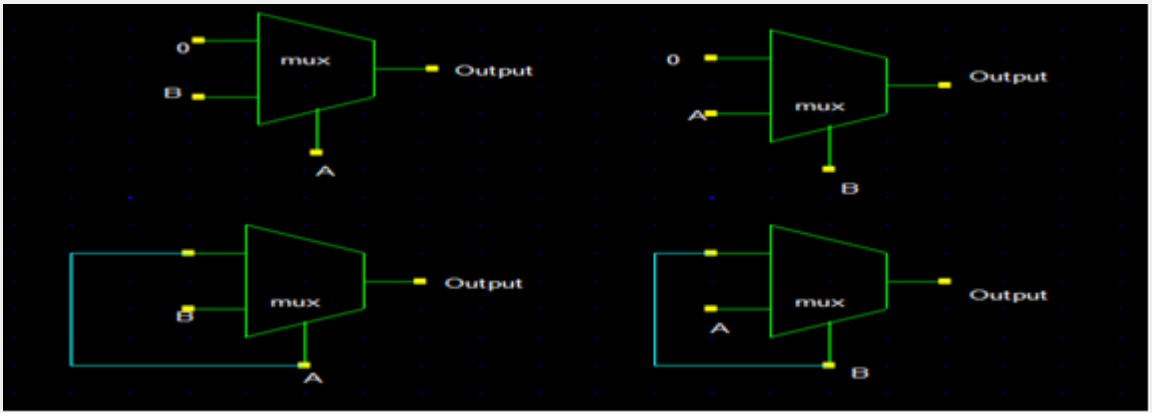
22. Design 4 bit comparator circuit?

Ans : Compares A = A3 A2 A1 A0 with B = B3 B2 B1 B0 Output = 1 if A = B



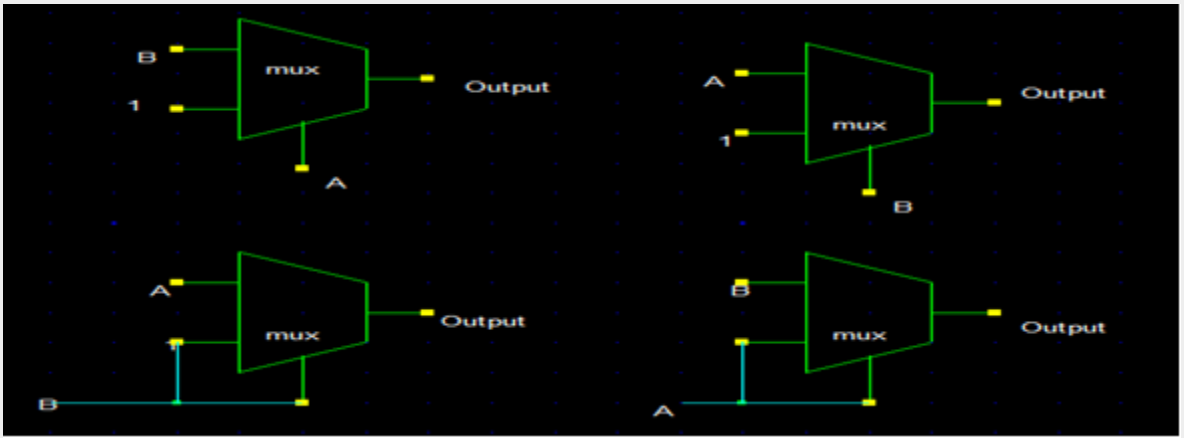
23. Design AND gate using Mux?

Ans :



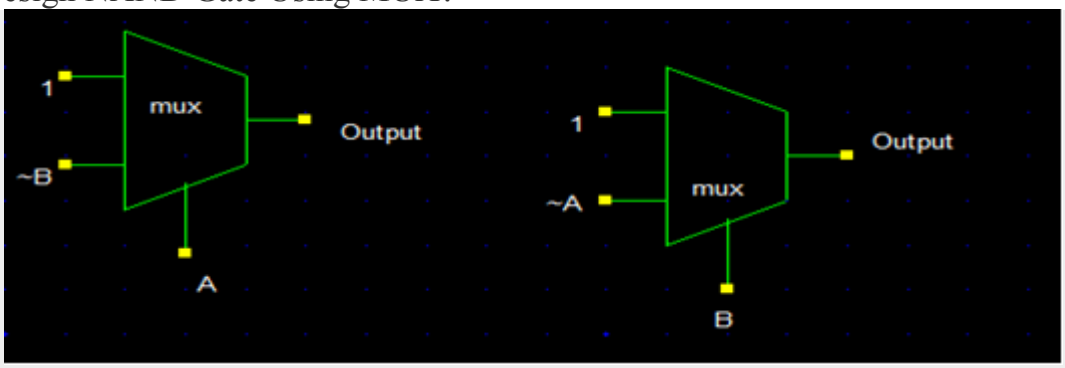
24. Design OR gate Using Mux?

Ans :



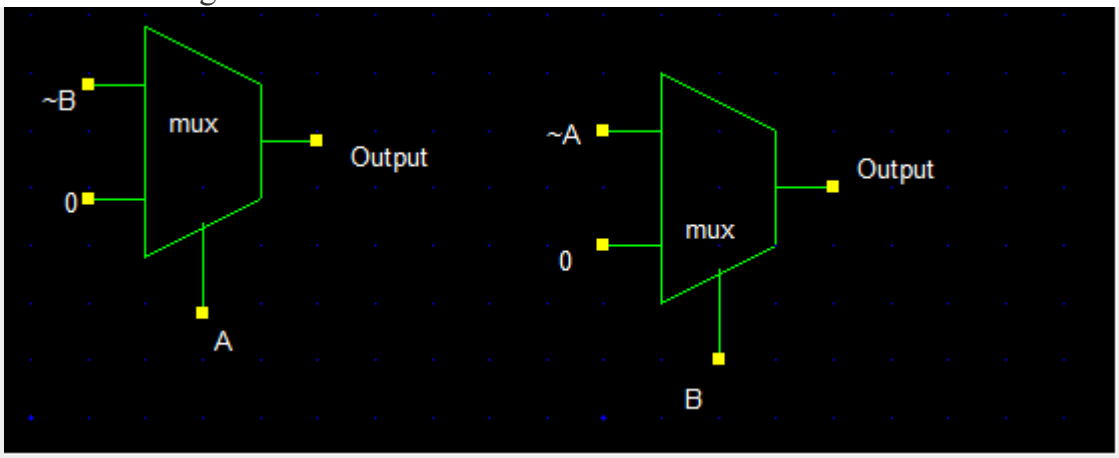
25. Design NAND Gate Using MUX?

Ans :



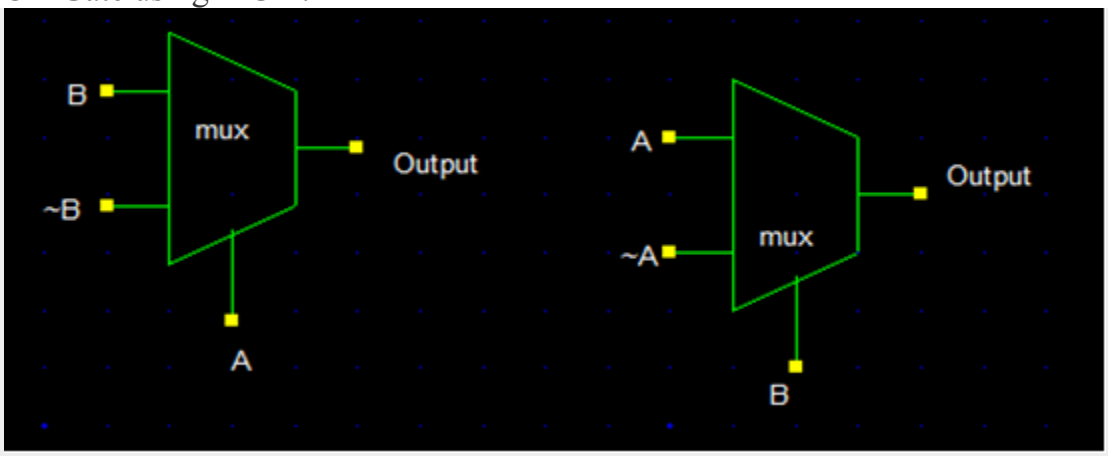
26. NOR Gate Using MUX?

Ans :



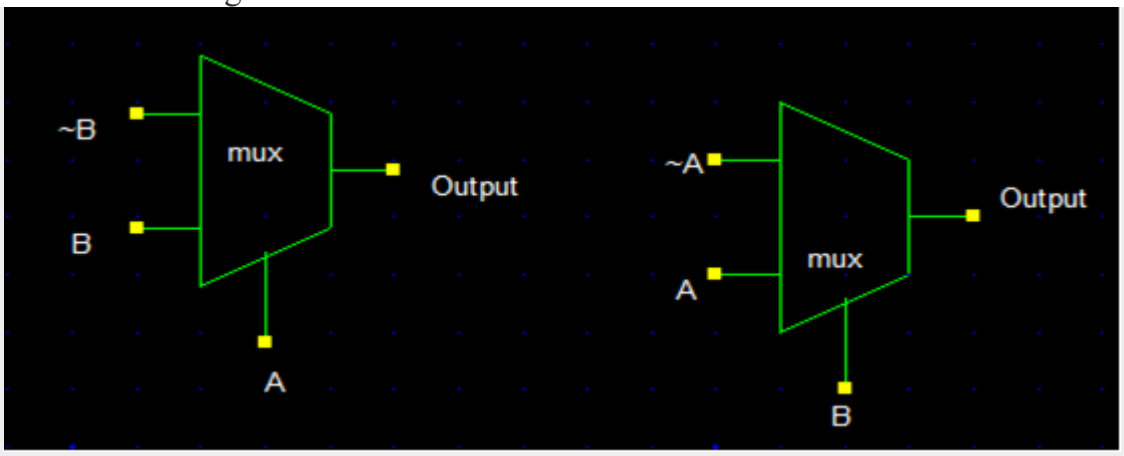
27. XOR Gate using MUX?

Ans :



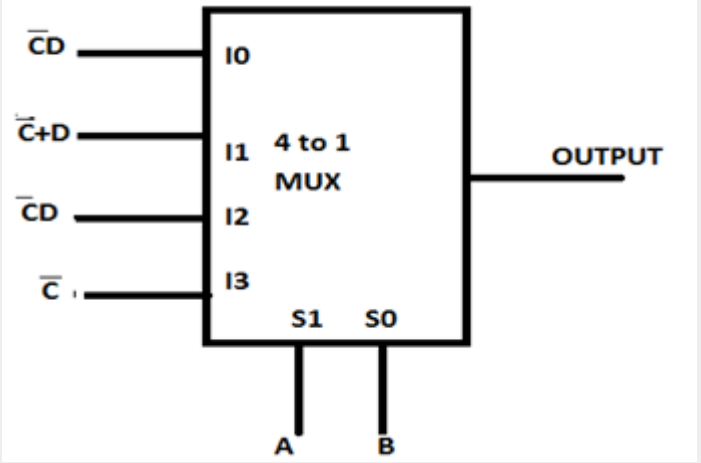
28. XNOR Gate Using MUX?

Ans :



29. Implement f (A,B,C,D) = ∑m(1,4, 5, 7, 9, 12, 13) using 4 to 1 mux?

Ans :



30. How to fix the setup and hold violation?

Ans : Setup violation can be fixed by slow down the clock. But hold time can't be fixed by this way. By inserting buffer in the path we canreduce the hold violations.

31. What is setup time and hold time?

Ans : Set up time is the amount of time before the clock edge that the input signal needs to be stable to guarantee it is accepted properly onthe clock edge. Hold time is the amount of time after the clock edge that same input signal has to be held before changing it to make sure it is sensed properly at the clock edge.

32. What will happen if there is setup time and hold time violation?

Ans : Enters into metastable state. At the end of metastable state, the flip-flop settles down to either '1' or '0'.

33. What is clock skew?

Ans : Clock skew is a phenomenon in synchronous circuits in which the clock signal (sent from the clock circuit) arrives at differentpoints(components) at different times.

34. Define local-skew, global-skew and useful-skew?

Ans : Local skew: The difference between the clock reaching at the launching flop v/s the clock reaching the destination flip-flop of atiming-path. Global skew: The difference between the earliest reaching flip-flop and latest reaching flip-flop for a same clock-domain. Useful skew: Useful skew is a concept of delaying the capturing flip-flop clock path, this approach helps in meeting setup requirement with in the launch and capture timing path. But the hold-requirement has to be met for the design.

35. What is meant by virtual clock? Why do i need it?

Ans : Virtual clock is mainly used to model the I/O timing specification. Based on what clock the input/output pads are passing the data.

36. What is slack?

Ans : 'Slack' is the amount of time that is measured from when an event 'actually happens' and when it must 'happen'. The term 'actually happens'can also be taken as being a predicted time for when the event will 'actually happen'. When something 'must happen' can also be called a 'deadline' so another definition of slack would be the time from when something 'actually happens' (call this Tact) until the deadline (call this Tdead).
Slack = Tdead – Tact.
Negative slack implies that the 'actually happen' time is later than the 'deadline' time.

37. Define Moore and Mealy state machine?

Ans : Mealy and Moore models are the basic models of state machines.
A state machine which uses only Entry Actions, so that its output depends on the state, is called a Moore model.
A state machine which uses only input actions, so that the output depends on the state and also on inputs, is called a Mealy model.

38. What is the difference between mealy and moore state machine?

Ans : Mealy machine has outputs that depend on the state and input. Moore machine has outputs that depend on only state. In Mealy as theoutput variable is a function both input and state, changes of state of the state variables will be delayed with respect to changes of signal level in the input variables, there are possibilities of glitches appearing in the output variables. Moore overcomes glitches as output dependent on only states and not the input signal level. The outputs are properties of states themselves. Which means that you get the output after the machine reaches a particular state, or to get some output your machine has to be taken to a state which provides you the output. Choice of a model depends on the application

39. Difference between one-hot and binary encoding?

Ans : A binary-encoded FSM design only requires as many flip-flops as are needed to uniquely encode the number of states in the state machine.The actual number of flip-flops required is equal to the ceiling of the log-base-2 of the number of states in the FSM. A one-hot FSM design requires a flip-flop for each state in the design and only one flipflop (the flip-flop representing the current or "hot" state) is set at a time in a one hot FSM design. For a state machine with 9- 16 states, a binary FSM only requires 4 flip-flops while a one-hot FSM requires a flip-flop for each state in the design.

40. What is significance of RAS and CAS in SDRAM?

Ans : SDRAM receives its address command in two address words. It uses a multiplex scheme to save input pins. The first address word islatched into the DRAM chip with the row address strobe. The RAS command is the column address strobe (CAS) for latching the second address word.

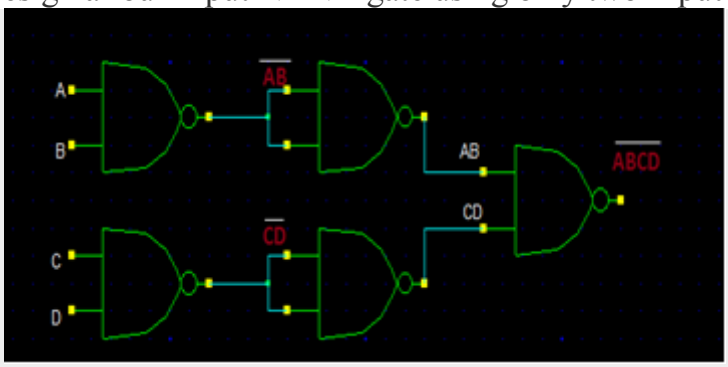
41. What will happen if contents of register are shifter left, right?

Ans : Ex: Consider 0110=6 a left shift will make it 1100=12, multiplied by 2 and right shift will make it 0011=3, right shift will Divide the value by 2.

42. For the FIFO rules, what are underflow and overflow conditions?

Ans : RULES: frequency(clk_A) = frequency(clk_B) / 4, period(en_B) = period(clk_A) * 100, duty_cycle(en_B) = 25%
Lets assume clk_B = 100MHz (10ns)
From (1), clk_A = 25MHz (40ns)
From (2), period(en_B) = 40ns * 400 = 4000ns, but we only output for 1000ns,due to (3), so 3000ns of the enable we are doing no output work. Therefore, FIFO size = 3000ns/40ns = 75 entries.

43. Design a four-input NAND gate using only two-input NAND gates.

Ans : 

44. Why interrupts are active low?

Ans : At transistor level of a module, active low means the capacitor in the output terminal gets charged or discharged based on low to high andhigh to low transition respectively. When it goes from high to low it depends on the pull down resistor that pulls it down and it is relatively easy for the output capacitance to discharge rather than charging.

45. Which one estimating maximum clock frequency of a circuit, set up time or hold time?

Ans : Setup time

46. Differences between D-Latch and D flip-flop?

Ans : D-latch is level sensitive where as flip-flop is edge sensitive. Flip-flops are made up of latches.

47. What is a multiplexer?

Ans : Multiplexer is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line.

48. How can you convert an SR Flip-flop to a JK Flip-flop?

Ans : By giving the feedback we can convert, i.e !Q=>S and Q=>R. Hence the S and R inputs will act as J and K respectively.

49. How can you convert the JK Flip-flop to a D Flip-flop?

Ans : By connecting the J input to the K through the inverter.

50. What is Race-around problem? How can you rectify it?
Ans : A condition in which, the clock pulse that remains in the 1 state while both J and K are equal to 1 will cause the output to complementagain and repeat complementing until the pulse goes back to 0. By using master-slave or edge-triggered flip-flop we can overcome this.

51. How do you detect if two 4-bit signals are same?
Ans : XOR each bits of A with B and the output of 4 xor gates are then given as input to 4-input nor gate. If output is 1 then A=B.

52. 7 bit ring counter's initial state is 0100010. After how many clock cycles will it return to the initial state?
Ans : 6 clock cycles

53. How to convert D flip-flop into divide by 2?
Ans : Connect Qbar to D and apply the clock at clock of DFF and take the O/P at Q.

54. What is the max clock frequency the circuit can handle, if T_setup= 6nS T_hold = 2nS T_propagation = 10nS ?
Ans : Max. Freq of operation is given by 1/ (propagation delay+setup time) = 1/16ns = 62.5 Mhz

55. If N number of XNOR gates are connected in series such that the N inputs are given to A0 & A1 of first XNOR gate and A2 & O/P of First XNOR to second XNOR gate and so on. Nth XNOR gates output is final output. How does this circuit work?
Ans : If N=Odd, the circuit acts as even parity detector. If N=Even, just the opposite, it will be Odd parity detector.

56. How will you implement a Full subtracter from a Full adder?
Ans : All the bits of subtrahend should be connected to the xor gate. Other input to the xor being one. The input carry bit to the full addershould be made 1. Then the full adder works like a full subtracter.

57. What is the difference between setup and hold time?
Ans : Setup violations are related to two edges of clock, we can vary the clock frequency to correct setup violation. But for hold time, we onlyconcerned with one edge and not basically depend on clock frequency.

58. In a 3-bit Johnson's counter what are the unused states?
Ans : 2(power n)-2n is used to find the unused states in Johnson counter. For 3-bit counter it is 8-6=2 unused states.

59. What is an LFSR?
Ans : LFSR is a linear feedback shift register where the input bit is driven by a linear function of the overall shift register value.

60. What is false path? How to determine this? What are the effects of false path in circuit?

Ans : False path is defined as, the paths in the circuit which are never exercised during normal circuit operation for any set of inputs. StaticTiming Analysis tools are able to identify simple false paths. However they are not able to identify all the false paths and sometimes report false paths as critical paths. Removal of false paths makes circuit testable and its timing performance predictable.

61. If two similar processors, one with a clock skew of 100ps and other with a clock skew of 60ps. Which one will consume more power? Why?

Ans : Clock skew of 60ps is more likely to have clock power. Because it is likely that low-skew processor has better designed clock tree withmore powerful and number of buffers and overheads to make skew better.

62. What are multi-cycle paths?

Ans : Multi-cycle paths are paths between registers that takes more than one clock cycle to become stable. Place and Route tools are capable offixing multi-cycle paths problem.

63. If two counters counting up to 16, built from negedge DFF, First circuit is synchronous and second is "ripple", which circuit has a less propagation delay? Why?

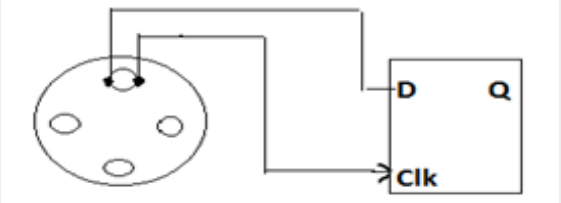
Ans : The synchronous counter will have lesser delay as the input to each flop is readily available before the clock edge. Eg: 16 state counter = 4bit counter = 4 Flip flops Let 15ns be the delay of each flop.
The worst case delay of ripple counter = 15 * 4 = 60ns
The delay of synchronous counter = 15ns

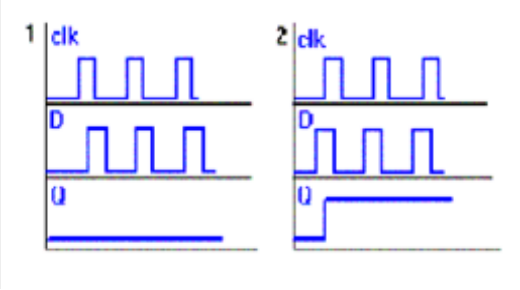
64. What is the difference between RAM and FIFO?

Ans : FIFO does not have address lines Ram is used for storage purpose where as fifo is used for synchronization purpose.

65. The circle can rotate clockwise and back. Use minimum hardware to build a circuit to indicate the direction of rotation?

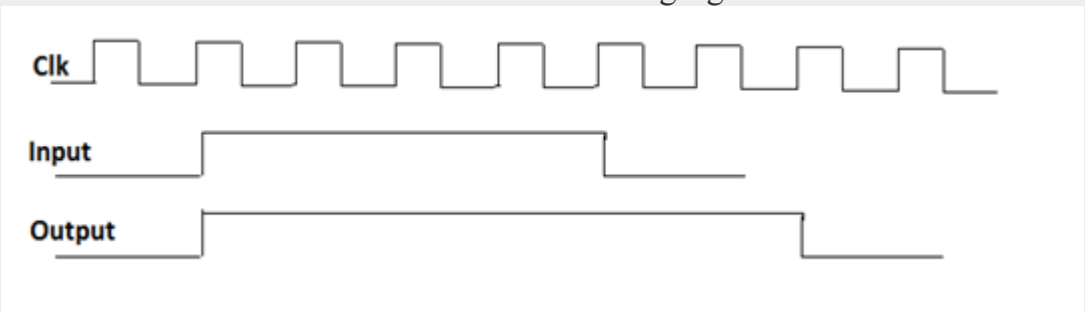
Ans : Two sensors are required to find out the direction of rotating. One of them is connected to the data input of D flip-flop and second oneto the clock input. If the circle rotates the way clock sensor sees the light first while D input (second sensor) is zero - the output of the flip-flop equals zero and if D input sensor "fires" first - the output of the flip-flop becomes high.

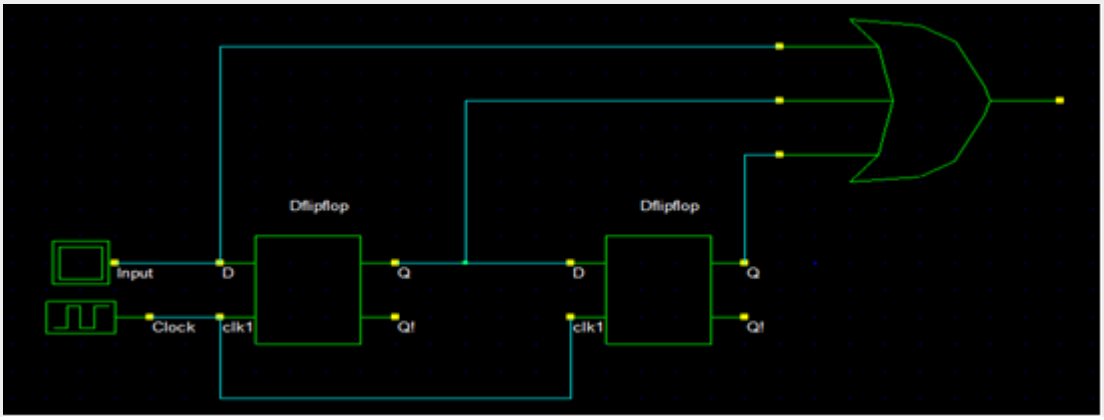




66. Give the circuit to extend the falling edge of the input by 2 clock pulses?

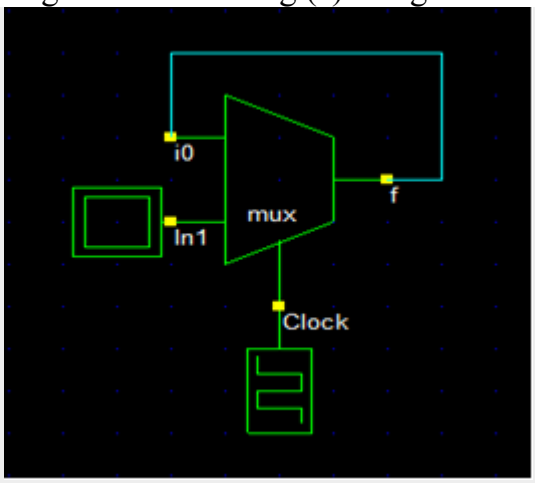
Ans : The waveforms are shown in the following figure.

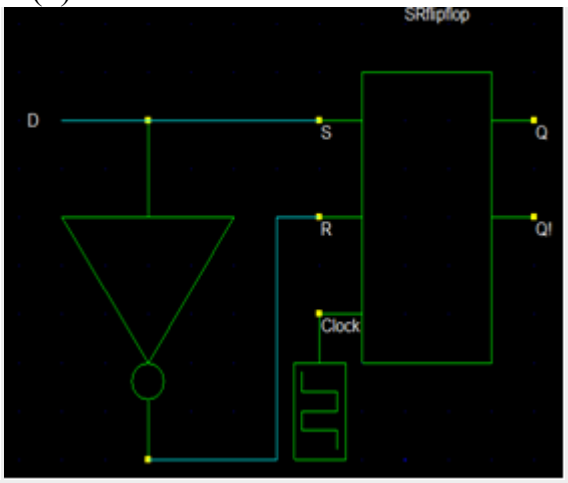




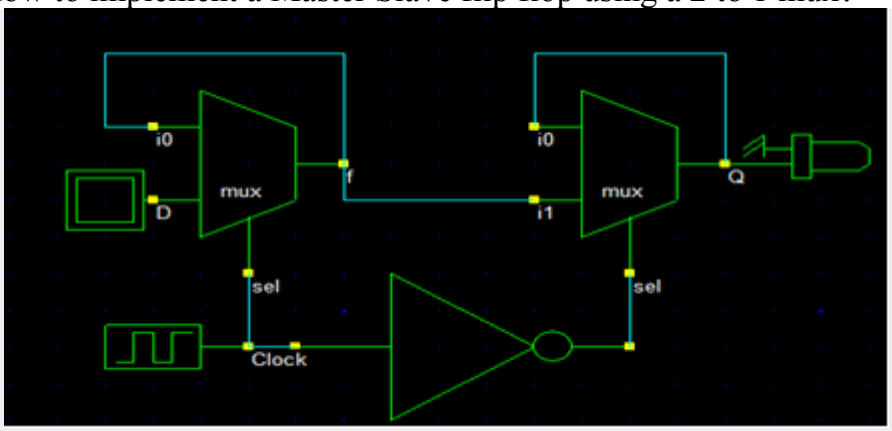
67. Design a D-latch using (a) using 2:1 Mux (b) from S-R Latch ?

Ans :





68. How to implement a Master Slave flip flop using a 2 to 1 mux?

Ans : 

69. How many 2 input xor's are needed to implement 16 input parity generator?

Ans : It is always n-1, where n is number of inputs. So 16 input parity generator will require 15 two input xor's.

70. Design a circuit for finding the 9's compliment of a BCD number using 4-bit binary adder and some external logic gates?

Ans : 9's compliment is nothing but subtracting the given no from 9. So using a 4 bit binary adder we can just subtract the given binary no from1001 here we can use the 2's compliment method addition.

71. What is the difference between write-back and write through cache?

Ans : Write-back caching method in which modifications to data in the cache are not copied to the cache source until absolutely necessary.Write-through cache performs all write operations in parallel, data is written to main memory and the L1 cache simultaneously. Write-back caching yields somewhat better performance than write-through caching because it reduces the number of write operations to main memory. A write-back cache is also called a copy-back cache.

72. What is the difference between Synchronous, Asynchronous communication?

Ans : Asynchronous systems do not send separate information to indicate the encoding or clocking information. The receiver must decide theclocking of the signal on it’s own. This means that the receiver must decide where to look in the signal stream to find ones and zeros, and decide for itself where each individual bit stops and starts. Synchronous systems negotiate the connection at the data-link level before communication begins. Basic synchronous systems will synchronize two clocks before transmission, and reset their numeric counters for errors etc. More advanced systems may negotiate things like error correction and compression. It refers to processes where data must be delivered within certain time constraints.